

CBSE Class 12 Chemistry — Alcohols, Phenols and Ethers

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (MCQs / Assertion-Reason / Very Short Answer)

Q1. The major organic product of the acid-catalysed dehydration of 1-methylcyclohexanol is: (a) 1-methylcyclohexane (b) 1-methylcyclohexene (c) 1-cyclohexylmethanol (d) 1-methylenecyclohexane [PYQ SQP 2022] [Confidence: Very High]

Q2. Which of the following reagents is most specifically used to distinguish between phenol and ethanol? (a) NaOH(aq.) (b) Neutral FeCl_3 (c) H_2/Ni (d) NaHCO_3 [PYQ Term-1 2021] [Confidence: Very High]

Q3. Which of the following aromatic acids reacts with acetic anhydride in the presence of an acid catalyst to form the compound Aspirin? (a) Benzoic acid (b) Salicylic acid (c) Phthalic acid (d) Acetic acid [PYQ SQP Term-1 2021] [Confidence: Very High]

Q4. The famous organic reaction specifically used to prepare salicylaldehyde from phenol using chloroform and aqueous NaOH is called: (a) Kolbe's reaction (b) Etard reaction (c) Reimer-Tiemann reaction (d) Stephen's reduction [PYQ SQP Term-1 2021] [Confidence: Very High]

Q5. When benzene diazonium chloride reacts with phenol in a weakly alkaline medium, it forms an orange dye. This reaction is chemically classified as a: (a) Diazotisation reaction (b) Condensation reaction (c) Coupling reaction (d) Acetylation reaction [PYQ 2023] [Confidence: Very High]

Q6. Predict the major product formed in the following oxidation reaction:

$\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_2 - \text{OH} \xrightarrow{\text{PCC}}$ (a) $\text{CH}_3 - \text{CHO}$ and $\text{CH}_3\text{CH}_2\text{OH}$ (b)

$\text{CH}_3 - \text{CH} = \text{CH} - \text{COOH}$ (c) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CHO}$ (d)

$\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CHO}$ [PYQ Term-1 2021] [Confidence: Very High]

Q7. During the dehydration of alcohols to alkenes by heating with concentrated H_2SO_4 , the crucial initiation step in the mechanism is: (a) Protonation of the alcohol molecule (b) Formation of a carbocation (c) Elimination of water (d) Formation of an ester [PYQ SQP Term-1 2021] [Confidence: High]

Q8. The $\text{C} - \text{O} - \text{H}$ bond angle in aliphatic alcohols is: (a) Slightly greater than $109^\circ 28'$ (b) Slightly less than $109^\circ 28'$ (c) Exactly 120° (d) Slightly less than 120° [PYQ Term-1 2021] [Confidence: High]

Q9. Williamson's synthesis of preparing dimethyl ether from sodium methoxide and methyl bromide primarily follows which reaction mechanism? (a) $\text{S}_\text{N}1$ (b) $\text{E}1$ (c) $\text{S}_\text{N}2$ (d) $\text{E}2$ [PYQ SQP Term-1 2021] [Confidence: Very High]

Q10. What is the common name of the alcohol $CH_3 - CH_2 - CH(OH) - CH_3$? (a) n-Butyl alcohol (b) sec-Butyl alcohol (c) iso-Butyl alcohol (d) tert-Butyl alcohol [PYQ Term-1 2021] [Confidence: High]

Q11. Assertion (A): An ether is more volatile than an alcohol of comparable molecular mass. Reason (R): Ethers are polar in nature. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [PYQ SQP 2022] [Confidence: Very High]

Q12. Assertion (A): Phenol is a weaker acid than p-nitrophenol. Reason (R): The $-NO_2$ group is an electron-withdrawing group and stabilizes the phenoxide ion. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. [Practice / PYQ Theme] [Confidence: Very High]

Q13. Which of the following isomeric nitrophenols is the most volatile with steam? (a) o-Nitrophenol (b) m-Nitrophenol (c) p-Nitrophenol (d) All are equally volatile [PYQ 2014] [Confidence: Very High]

Q14. Arrange the following in the strictly increasing order of their boiling points: Propanol, Propane, Propanal. [PYQ 2017] [Confidence: Very High]

Q15. Write the structure of the main product formed when anisole ($C_6H_5 - O - CH_3$) is treated with a mixture of concentrated HNO_3 and concentrated H_2SO_4 . [Practice] [Confidence: Medium]

Q16. Identify the major products **A** and **B** formed in the cleavage reaction:
 $C_6H_5 - CH_2 - O - C_6H_5 + HBr \rightarrow A + B$. [PYQ SQP 2022] [Confidence: High]

Q17. Predict the exact reagent needed for carrying out the conversion of Phenol to benzoquinone. [PYQ 2020] [Confidence: High]

Q18. Which specific alkyl halide is formed exclusively when tert-butyl methyl ether, $(CH_3)_3C - O - CH_3$, is heated with **HI**? [PYQ 2015] [Confidence: Very High]

Q19. Does the Lucas reagent ($ZnCl_2$ / conc. HCl) react fastest to produce turbidity with primary, secondary, or tertiary alcohols? [Practice] [Confidence: Very High]

Q20. Name the specific enzyme which catalyses the conversion of glucose or fructose directly into ethanol. [Practice] [Confidence: Medium]

2-Mark Questions (Short Answer)

Q21. Write the complete step-by-step mechanism for the hydration of ethene to yield ethanol in the presence of an acid catalyst. [PYQ 2020] [Confidence: Very High]

Q22. Give structural reasons for the following observation: Ortho-nitrophenol is steam volatile, while para-nitrophenol is not steam volatile. [PYQ 2023] [Confidence: Very High]

- Q23.** Write the balanced chemical equations involved in the following important organic reactions: (i) Kolbe's reaction (ii) Williamson synthesis [PYQ 2023] [Confidence: Very High]
- Q24.** Carry out the following specific conversion in no more than two steps: Phenol to salicylaldehyde. [PYQ 2020] [Confidence: High]
- Q25.** How do you chemically convert Phenol into 2,4,6-tribromophenol? Write the balanced chemical equation. [PYQ 2020] [Confidence: Very High]
- Q26.** Out of t-butyl alcohol and n-butanol, which one will undergo acid-catalysed dehydration much faster to form an alkene, and why? [PYQ 2020] [Confidence: Very High]
- Q27.** Give simple, distinctly observable chemical tests to distinguish between the following pairs of compounds: (i) Ethanol and phenol (ii) Propan-1-ol and 2-methylpropan-2-ol [PYQ 2017] [Confidence: Very High]
- Q28.** Explain the following observation: The boiling point of ethanol is significantly higher than that of its isomer, methoxymethane. [PYQ 2015] [Confidence: Very High]
- Q29.** Write the structures of the major organic products formed when butan-2-ol reacts with the following: (i) CrO_3 (ii) SOCl_2 [PYQ 2017] [Confidence: High]
- Q30.** How are the following conversions neatly carried out? (i) Propene to propan-2-ol (ii) Benzyl chloride to benzyl alcohol [PYQ 2015] [Confidence: High]
- Q31.** Give a detailed chemical reason for the following statement: Phenol is noticeably more acidic than ethanol. [PYQ 2017] [Confidence: Very High]
- Q32.** Write the equation involved in the Friedel-Crafts alkylation reaction of anisole with chloromethane in the presence of anhydrous AlCl_3 . [PYQ 2019] [Confidence: High]
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3-Mark Questions (Short Answer)

- Q33.** Write the detailed mechanism (using curved arrow notation) of the acid-catalysed dehydration of ethanol to yield ethene at 443 K. [PYQ 2018] [Confidence: Very High]
- Q34.** Write the detailed mechanism of the acid-catalysed dehydration of ethanol to yield ethoxyethane (diethyl ether) at the controlled temperature of 413 K. [PYQ 2023] [Confidence: Very High]
- Q35.** How will you successfully carry out the following conversions in organic chemistry? (i) Phenol to anisole (ii) Propan-2-ol to 2-methylpropan-2-ol (iii) Aniline to phenol [PYQ 2015] [Confidence: High]
- Q36.** Give the exact structures of the final major products expected from the following reactions: (i) Hydroboration of propene followed by oxidation with H_2O_2 in an alkaline medium. (ii) Dehydration of $(\text{CH}_3)_3\text{C} - \text{OH}$ by gently heating it with 20% H_3PO_4 at 358 K. (iii) Heating of anisole with an excess of HI . [PYQ 2020] [Confidence: Very High]
- Q37.** Account chemically for the following strict observations: (i) The preparation of ethers by acid dehydration of secondary or tertiary alcohols is not a suitable method. (ii) The $\text{C} - \text{O}$

bond length in phenol is shorter than the $C - O$ bond length in methanol. (iii) Lower aliphatic alcohols are highly soluble in water. [Practice / PYQ Theme] [Confidence: Very High]

Q38. Write the main organic product(s) in each of the following chemical reactions: (i)

$C_6H_5 - OH + aq. NaOH \xrightarrow{CO_2, H^+} ?$ (ii) $CH_3 - CH = CH_2 \xrightarrow{H_2O/H^+} ?$ (iii) Phenol is strongly heated with Zinc dust. [PYQ 2016 / Practice] [Confidence: High]

Q39. Give reliable, simple chemical tests to uniquely distinguish between the following pairs:

(i) Methanol and Ethanol (ii) Phenol and Benzoic acid (iii) Propan-1-ol and Propan-2-ol

[Practice] [Confidence: Very High]

Q40. Why does the ether $(CH_3)_3C - O - CH_3$ on reaction with HI strictly give CH_3OH and $(CH_3)_3C - I$ as the main products, and not $(CH_3)_3C - OH$ and CH_3I ? Explain the exact reasoning or mechanism governing this. [PYQ 2015] [Confidence: Very High]

Q41. Predict the major products of the following electrophilic substitution reactions on phenol: (i) Phenol reacts with Br_2 in CS_2 at 273 K. (ii) Phenol reacts with concentrated HNO_3 . (iii) Phenol reacts with dilute HNO_3 at room temperature. [Practice] [Confidence: High]

Q42. Write the names of the specific reagents required and the balanced equations for: (i) Oxidation of a primary alcohol to a carboxylic acid. (ii) Oxidation of a primary alcohol to an aldehyde (without further oxidation). (iii) Nitration of phenol to yield picric acid. [Practice] [Confidence: Medium]

5-Mark Questions (Long Answer / Case-Study)

Q43. (A) Arrange the following specific sets of compounds in the increasing order of their indicated property: (i) p-nitrophenol, ethanol, phenol (Increasing acidic character) (ii) Propanol, Propane, Propanal (Increasing boiling point) (2 Marks) (B) Write the mechanism of the reaction of HI with methoxymethane $(CH_3 - O - CH_3)$. (2 Marks) (C) Write the exact IUPAC name of the compound: $CH_3 - CH(OH) - CH_2 - C(CH_3)_3$. (1 Mark) [PYQ 2017 / Practice] [Confidence: Very High]

Q44. An organic compound **A** having the molecular formula C_6H_6O gives a characteristic violet colour with neutral $FeCl_3$ solution. When compound **A** is treated with $NaOH$ and CO_2 at 400 K under pressure, it gives a compound **B**, which upon acidification forms compound **C**. Compound **C** reacts rapidly with acetyl chloride to form a widely used painkiller **D**. (i) Identify the exact structures and names of **A**, **B**, **C**, and **D**. (4 Marks) (ii) Name the specific chemical reaction involved in the conversion of **A** to **B** & **C**. (1 Mark) [Practice based on CBSE PYQ trends] [Confidence: Very High]

Q45. (A) Explain precisely why Williamson's ether synthesis is strictly successful with primary alkyl halides but completely fails with tertiary alkyl halides to yield ethers. Give chemical reactions to support your answer using sodium methoxide and tert-butyl bromide. (3 Marks) (B) Write the necessary chemical equations to show how the following are commercially

obtained: (i) Phenol from Cumene. (ii) Anisole from phenol. (2 Marks) [PYQ 2020 / Practice]
[Confidence: High]

Q46. Write the detailed, balanced chemical equations to appropriately illustrate the following important organic name reactions: (i) Reimer-Tiemann reaction (ii) Kolbe's reaction (iii) Williamson's synthesis (iv) Hydroboration-oxidation of an alkene (v) Catalytic dehydrogenation of primary alcohols using heated Cu. (1 Mark each = 5 Marks) [PYQ 2014, 2015, 2023] [Confidence: Very High]

Q47. Case Study: Acidity of Alcohols and Phenols Alcohols and phenols are acidic in nature because the $O-H$ bond is polar. Electron-withdrawing groups attached to the phenol ring heavily increase its acidic strength by stabilizing the phenoxide ion through delocalization of negative charge, while electron-donating groups decrease the acidity. Furthermore, phenols are significantly more acidic than aliphatic alcohols. (i) Between p-nitrophenol and p-cresol (4-methylphenol), which is significantly more acidic and why? (2 Marks) (ii) Why are primary alcohols generally considered weaker acids than water? (1 Mark) (iii) Briefly explain the mechanism of nucleophilic attack of an alcohol on a protonated carboxylic acid during the esterification process. (2 Marks) [Practice] [Confidence: High]

Q48. (A) Explain how you will securely distinguish between primary, secondary, and tertiary alcohols using the Lucas test. Detail the observations and the chemistry behind it. (3 Marks)
(B) Provide the balanced chemical equations for the following conversions: (i) Ethanol to acetone (Propanone). (ii) Benzene to phenol. (2 Marks) [Practice / PYQ themes] [Confidence: High]

Q49. (A) Carefully complete the following chemical reactions: (i)
 $CH_3-CH_2-O-CH_2-CH_3 + PCl_5 \rightarrow ?$ (ii)
 $C_6H_5-O-CH_3 + HBr \text{ (excess)} \rightarrow ?$ (iii) $CH_3-CH_2-OH \xrightarrow{\text{conc. } H_2SO_4, 443K} ?$ (3 Marks)
(B) Write the skeletal structure of 2-ethoxy-3-methylpentane. (1 Mark) (C) Name the best specific reagent used to selectively convert ethanol to ethanoic acid. (1 Mark) [Practice] [Confidence: Medium]

Q50. (A) A compound **X** (C_2H_6O) on oxidation by passing over heated **Cu** at 573 K gives **Y** (C_2H_4O). Compound **Y** successfully reduces Tollens' reagent to give a silver mirror. Compound **X** on severe dehydration at 443 K with concentrated H_2SO_4 yields a gas **Z** (C_2H_4). Identify the exact structures and names of **X**, **Y**, **Z** and write the involved reactions. (3 Marks)
(B) Why do ethers reliably possess a net dipole moment even if they are structurally symmetrical (e.g., diethyl ether)? (2 Marks) [PYQ 2016 / Practice] [Confidence: Very High]

Answer Key

Q1. (b) 1-methylcyclohexene. (Follows Saytzeff's rule to give the more substituted, stable alkene). **Q2.** (b) Neutral $FeCl_3$. (Phenol gives a violet colour complex, ethanol gives no colour). **Q3.** (b) Salicylic acid. (Reacts with acetic anhydride to form acetylsalicylic acid/Aspirin).

Q4. (c) Reimer-Tiemann reaction. **Q5.** (c) Coupling reaction. (Yields p-hydroxyazobenzene, an orange dye). **Q6.** (c) $CH_3 - CH = CH - CHO$. (PCC oxidizes primary alcohols to aldehydes without affecting $C = C$ double bonds). **Q7.** (a) Protonation of the alcohol molecule (to make $-OH_2^+$, a better leaving group). **Q8.** (b) Slightly less than $109^\circ 28'$. (Due to lone pair-lone pair repulsion on the oxygen atom). **Q9.** (c) S_N2 . (Nucleophilic substitution bimolecular). **Q10.** (b) sec-Butyl alcohol. **Q11.** (b) Both A and R are true but R is NOT the correct explanation of A. (Volatility is due to the lack of intermolecular hydrogen bonding in ethers, not just polarity). **Q12.** (a) Both A and R are true and R is the correct explanation of A. **Q13.** (a) o-Nitrophenol. (Due to intramolecular hydrogen bonding; others have intermolecular H-bonding causing higher boiling points). **Q14.** Propane < Propanal < Propanol. **Q15.** A mixture of 2-nitroanisole (minor) and 4-nitroanisole (major). **Q16.** A = Phenol (C_6H_5OH), B = Benzyl bromide ($C_6H_5CH_2Br$). (The $O - C_6H_5$ bond does not break due to partial double bond character). **Q17.** $Na_2Cr_2O_7$ / conc. H_2SO_4 (Acidified sodium dichromate). **Q18.** tert-Butyl iodide ($(CH_3)_3C - I$). **Q19.** Tertiary alcohols. (Produces immediate turbidity via a stable 3° carbocation). **Q20.** Zymase.

Q21. Mechanism involves 3 steps: 1. Protonation of alkene to form carbocation: $CH_2 = CH_2 + H_3O^+ \rightleftharpoons CH_3 - CH_2^+ + H_2O$. 2. Nucleophilic attack of water: $CH_3 - CH_2^+ + H_2O \rightleftharpoons CH_3 - CH_2 - OH_2^+$. 3. Deprotonation: $CH_3 - CH_2 - OH_2^+ + H_2O \rightleftharpoons CH_3 - CH_2 - OH + H_3O^+$. **Q22.** o-Nitrophenol has intramolecular hydrogen bonding, making it exist as discrete molecules (steam volatile). p-Nitrophenol has intermolecular hydrogen bonding, causing molecular association and a higher boiling point. **Q23.** (i) Phenol $\xrightarrow{NaOH}$ Sodium phenoxide $\xrightarrow{CO_2, H^+}$ Salicylic acid. (ii) $R - X + R' - O^- Na^+ \rightarrow R - O - R' + NaX$. **Q24.** Treat Phenol with $CHCl_3$ and aq. $NaOH$ at 340K, followed by acidification (Reimer-Tiemann reaction). **Q25.** $C_6H_5OH + 3Br_2(aq) \rightarrow$ 2,4,6-tribromophenol (white ppt) + $3HBr$. **Q26.** t-butyl alcohol reacts much faster because it forms a stable 3° carbocation intermediate. n-butanol forms a highly unstable 1° carbocation. **Q27.** (i) Add neutral $FeCl_3$; phenol gives violet colour, ethanol does not. (ii) Lucas test: 2-methylpropan-2-ol gives immediate turbidity; propan-1-ol gives none at room temperature. **Q28.** Ethanol undergoes extensive intermolecular hydrogen bonding, requiring more energy to break bonds. Ethers only have weak dipole-dipole interactions. **Q29.** (i) Butanone ($CH_3 - CO - CH_2 - CH_3$). (ii) 2-Chlorobutane ($CH_3 - CH(Cl) - CH_2 - CH_3$) + $SO_2 \uparrow$ + $HCl \uparrow$. **Q30.** (i) Add H_2O/H^+ (Markovnikov hydration). (ii) Boil with aqueous $NaOH$ or KOH . **Q31.** Phenol loses H^+ to form a phenoxide ion which is heavily stabilized by resonance. Ethanol forms an ethoxide ion that has no resonance stabilization. **Q32.** Anisole + $CH_3Cl \xrightarrow{\text{Anhydrous } AlCl_3} 2\text{-methoxytoluene (minor)} + 4\text{-methoxytoluene (major)} + HCl$.

Q33. 1. Protonation of alcohol: $CH_3CH_2OH + H^+ \rightleftharpoons CH_3CH_2OH_2^+$. 2. Formation of carbocation (slow): $CH_3CH_2OH_2^+ \rightarrow CH_3CH_2^+ + H_2O$. 3. Elimination of proton to form alkene: $CH_3CH_2^+ \rightarrow CH_2 = CH_2 + H^+$. **Q34.** 1. Protonation: $CH_3CH_2OH + H^+ \rightleftharpoons CH_3CH_2OH_2^+$. 2. Nucleophilic attack by second alcohol molecule (S_N2): $CH_3CH_2OH + CH_3CH_2OH_2^+ \rightarrow CH_3CH_2 - O^+(H) - CH_2CH_3 + H_2O$. 3.

Deprotonation: loss of H^+ yields ethoxyethane. **Q35.** (i) Phenol $\xrightarrow{NaOH}$ Sodium phenoxide $\xrightarrow{CH_3I}$ Anisole. (ii) Propan-2-ol $\xrightarrow{CrO_3}$ Propanone $\xrightarrow{CH_3MgBr, H_3O^+}$ 2-methylpropan-2-ol. (iii) Aniline $\xrightarrow{NaNO_2/HCl, 0^\circ C}$ Benzene diazonium chloride $\xrightarrow{H_2O, \Delta}$ Phenol. **Q36.** (i) Propan-1-ol. (Anti-Markovnikov hydration). (ii) 2-Methylpropene. (iii) Phenol (C_6H_5OH) and Methyl iodide (CH_3I). **Q37.** (i) 2° and 3° alcohols readily undergo elimination competing with substitution, heavily yielding alkenes instead of ethers. (ii) In phenol, the oxygen lone pair resonates with the benzene ring, giving the $C - O$ bond partial double bond character. (iii) They can form hydrogen bonds with water molecules. **Q38.** (i) Salicylic acid (2-hydroxybenzoic acid). (ii) Propan-2-ol. (iii) Benzene. **Q39.** (i) Iodoform test: Ethanol gives a yellow precipitate of CHI_3 , methanol does not. (ii) $NaHCO_3$ test: Benzoic acid gives brisk effervescence of CO_2 , phenol does not. (iii) Iodoform test: Propan-2-ol gives yellow ppt of CHI_3 ; propan-1-ol does not. **Q40.** The reaction proceeds via an S_N1 mechanism. The ether oxygen is protonated, and then the bulky $(CH_3)_3C^+$ carbocation forms because it is highly stable (3°). The I^- then exclusively attacks this 3° carbocation forming $(CH_3)_3C - I$. **Q41.** (i) 2-bromophenol and 4-bromophenol (major). (ii) 2,4,6-trinitrophenol (Picric acid). (iii) o-nitrophenol and p-nitrophenol. **Q42.** (i) Acidified $KMnO_4$ or $K_2Cr_2O_7/H_2SO_4$. (ii) Pyridinium chlorochromate (PCC) or CrO_3 in anhydrous medium. (iii) Conc. HNO_3 in conc. H_2SO_4 . **Q43.** (A) (i) Ethanol < Phenol < p-nitrophenol. (ii) Propane < Propanal < Propanol. (B) Mechanism: Protonation gives $CH_3 - O^+H - CH_3$. I^- attacks via S_N2 , breaking the bond to yield CH_3I and CH_3OH . (C) 4,4-dimethylpentan-2-ol. **Q44.** (i) A = Phenol, B = Sodium salicylate, C = Salicylic acid, D = Aspirin (2-acetoxybenzoic acid). (ii) Kolbe's reaction. **Q45.** (A) Alkoxide is a strong base as well as a nucleophile. With a 3° alkyl halide, steric hindrance prevents S_N2 substitution; instead, rapid E2 elimination occurs, forming an alkene. Reaction: $(CH_3)_3C - Br + CH_3ONa \rightarrow (CH_3)_2C = CH_2 + NaBr + CH_3OH$. (B) (i) Cumene is oxidized with O_2 to cumene hydroperoxide, which on acid hydrolysis gives phenol + acetone. (ii) Phenol + $NaOH \rightarrow$ Sodium phenoxide. Then, treat with $CH_3Cl \rightarrow$ Anisole + $NaCl$. **Q46.** (i) Phenol + $CHCl_3 + 3NaOH \xrightarrow{340K}$ Salicylaldehyde. (ii) Phenol + $NaOH \rightarrow$ Sod. Phenoxide $\xrightarrow{CO_2, 400K}$ Salicylic acid. (iii) $R - X + R'ONa \rightarrow R - O - R' + NaX$. (iv) $3CH_3CH = CH_2 + BH_3 \rightarrow (CH_3CH_2CH_2)_3B \xrightarrow{H_2O_2/OH^-}$ $3CH_3CH_2CH_2OH$. (v) $R - CH_2OH \xrightarrow{Cu, 573K}$ $R - CHO + H_2$. **Q47.** (i) p-nitrophenol. The $-NO_2$ group is strongly electron-withdrawing (-R, -I), stabilizing the phenoxide ion, while $-CH_3$ in p-cresol is electron-donating (+I, hyperconjugation), destabilizing it. (ii) The alkyl group in alcohols is electron-donating (+I effect), which increases electron density on oxygen, destabilizing the alkoxide ion compared to hydroxide. (iii) Protonation of the carbonyl oxygen of the acid increases electrophilicity of the carbonyl carbon. The alcohol's oxygen lone pair then attacks the carbonyl carbon to form a tetrahedral intermediate. **Q48.** (A) Add Lucas reagent ($ZnCl_2 +$ conc. HCl). 3° alcohol: Immediate turbidity (fast S_N1). 2° alcohol: Turbidity in 5 mins. 1°

alcohol: No turbidity at room temperature. (B) (i) Ethanol $\xrightarrow{PCC}$ Ethanal $\xrightarrow{CH_3MgBr, H_2O}$ Propan-2-ol $\xrightarrow{CrO_3}$ Acetone. (ii) Benzene $\xrightarrow{Cl_2, FeCl_3}$ Chlorobenzene $\xrightarrow{NaOH, 623K, 300atm}$ Sod. phenoxide $\xrightarrow{H^+}$ Phenol. **Q49.** (A) (i) $2CH_3CH_2Cl + POCl_3 + H_2O$. (ii) Phenol + Benzyl bromide (wait, if excess HBr, anisole $C_6H_5OCH_3$ cleaves to Phenol + CH_3Br . Phenol doesn't react further). (iii) Ethene ($CH_2 = CH_2$). (B) $CH_3 - CH(O - CH_2 - CH_3) - CH(CH_3) - CH_2 - CH_3$. (C) Acidified Potassium dichromate ($K_2Cr_2O_7/H_2SO_4$). **Q50.** (A) X is Ethanol (CH_3CH_2OH), Y is Ethanal (CH_3CHO), Z is Ethene ($CH_2 = CH_2$). (B) The $C - O - C$ bond is bent (V-shaped) due to the sp^3 hybridization of oxygen and two lone pairs repelling the bond pairs. Thus, the two $C - O$ bond dipole moments do not cancel each other out.

CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).